ITEC 116 Midterm Reviewer

Systems Integration and Architecture 2. Part 1 is the cheat sheet, Part 2 is the practice quiz. Tap "Show answer" to check yourself.

Part 1. Cheat sheet

Core definitions

- **ERP**: the digital backbone of an organization; unifies finance, HR, supply chain and customer service in one environment.
- **System integration planning**: combining discrete sub-systems into a unified infrastructure to guarantee seamless communication.
- **Logical integration** (human/business level): secure data sharing among stakeholders based on role authorization and operational need; needs organizational change and redefined roles.
- **Physical integration** (technical level): connectivity between heterogeneous applications, platforms, OSs and DBMSs.
- **Middleware**: software that presents a unified data view to end-users while keeping data integrity and synchronization across disparate back-end systems.
- **BPR** (Business Process Re-engineering): addresses the "people barrier" and shifts focus from departmental targets to organizational goals.
- **Data identification**: first step of data mapping; pinpoint the exact records needed.

ERP architecture generations

Gen	Era	Key traits
1st	1970s-1980s	Legacy mainframe silos; batch processing; point-to-point
2nd	1990s-early 2000s	Client-server; 2/3-tier; custom middleware or EAI
3rd	2010s-present	Web/cloud; multi-tenant; microservices; REST/SOAP; SOA

REST vs. SOAP

	REST	SOAP
Style	Lightweight, stateless, HTTP methods (GET, POST, PUT, DELETE)	Highly structured, protocol-independent

Format	Usually JSON	XML only, WSDL contracts
Security	HTTPS/TLS, OAuth 2.0, API keys; validation built into endpoint logic	WS-Security: message-level encryption, digital signatures; WS-AtomicTransaction
Best for	Real-time web apps, microservices	High-security banking and ERP transactions

Data transformation rules

- **Data type casting**: e.g., string date to epoch timestamp.
- **Value mapping**: e.g., CRM "Pending_Approval" to ERP "Code_02".
- **Aggregation and structuring**: nest line items into one JSON/XML array.

The 5 key planning questions

1. **Transactions**: ACID compliance or multi-system rollback on failure.
2. **Connectivity**: protocols (REST, SOAP, SFTP, AMQP) and styles (point-to-point, middleware/ESB, event-driven).
3. **Interfaces and data formatting**: payloads and transformation rules.
4. **Security and compliance**: authentication, encryption in transit/at rest, data protection standards.
5. **Error handling and resiliency**: logging, retries, admin alerts.

Case study: a custom endpoint (account + order in one request, with rollback) reduces network overhead and transaction risk (no orphan data) but costs more upfront development time than standard CRUD.

The 8 steps in system integration

1. **Resource categorization**: audit hardware, software, OS platforms, IS architectures.
2. **Compliance and standards**: verify support for universal standards (JDBC/ODBC).
3. **Legacy systems support**: migration and interface policies; encapsulate legacy assets.
4. **Middleware selection**: interim and long-term communication between legacy and modern systems.
5. **Authentication and authorization**: Single Sign-On (SSO), unified access policies.
6. **Centralized IT services**: central support and help desk.
7. **Backup, recovery and security**: disaster recovery, redundancy, security policies.
8. **Standardization policies**: procurement guidelines for future purchases.

Memory trick: Resources, Compliance, Legacy, Middleware, Authentication, Centralized, Backup, Standards.

Benefits vs. limitations of integration

Benefits	Limitations
Increased revenue and growth (lower operating and inventory costs) Competitive parity (SMEs join B2B supply chains) Enhanced information visibility (real-time decisions) Hardware/software standardization	High initial capital outlay Power and interdepartmental conflicts Long-term/intangible ROI Creativity constraints from strict standardization

PMO, scope and change control

- **PMO**: ensures cross-functional teams work cohesively and resolve functionality issues in a timely, open and efficient manner.
- **Scope**: the precise deliverables of a project. **Scope creep**: unmanaged, continuous changes to the original project parameters; hurts **Cost, Time, Quality**.
- Prevent scope creep by defining parameters in a **Business Case, Project Charter or Mission Statement**.
- **Change Control Board (CCB)**: senior management and key users who review and authorize scope changes.
- **Decision-making governance**: defines how and when decisions are finalized; without it, trade-offs are not evaluated and user buy-in is lost.

Change control white paper (5 parts):

1. Issue identification
2. Technical and user implications
3. Options and trade-offs
4. Resource impact (budget, staff hours, timeline)
5. Final recommendation (to the CCB for sign-off)

Teamwork and organizational change

- **Internal IT**: institutional knowledge, may lack ERP/middleware experience. **External consultants**: experience from other rollouts. **Vendor specialists**: bridge technical gaps.
- **Knowledge transfer goal**: consultants transfer knowledge to internal IT before offboarding.
- **Change management**: (1) transparent communication, (2) aggressive training before go-live, (3) executive sponsorship.

SDLC vs. ERPLC

SDLC	ERPLC
Determine requirements and build custom software from scratch	Modify business processes to fit prepackaged enterprise software (industry best practices)

Customizing core ERP is expensive, raises vendor support fees and complicates upgrades, so prioritize **BPR over code customization**.

Implementation strategies

Strategy	Description	Cost/Risk
Vanilla	Out-of-the-box core functionality; adapt processes to software	Lowest cost, BPR and time
Middle-of-the-Road	Moderate core changes plus significant BPR	Balanced
Comprehensive	Full functionality, industry modules, extensive BPR, heavy customization	Most expensive, longest, highest risk

5 stages of the ERP life cycle

1. **Scope and commitment**: business case, feasibility, scope, team, top-management commitment.
2. **Analysis and design**: gap analysis, change management plan, sandbox/prototype.
3. **Acquisition and development**: buy licenses, build infrastructure, configure security/roles, data migration mapping, preliminary training.
4. **Implementation (go-live)**: convert from legacy, release to users, help desk monitoring.
5. **Operation**: knowledge transfer, ongoing training, contract management, vendor patches.

4 cutover (conversion) strategies

Method	Description	Cost/Risk
Direct (Big Bang)	Immediate switch to new system	High risk, lowest upfront cost, fastest
Parallel	Legacy and new run together for a set period	Highest cost and workload, lowest risk
Phased	Gradual migration by department or module	Minimizes disruption
Pilot	Full system to a small group or one location first	Tests suitability before full rollout

Part 2. Practice quiz

Matches the TOS: Knowledge and Comprehension (identification, multiple choice), Analysis (true

or false), Synthesis (essay).

A. Identification

1. The process of combining discrete sub-systems into a unified infrastructure to guarantee seamless communication.

▶ Show answer

2. Integration that focuses on secure data sharing among stakeholders based on role authorization.

▶ Show answer

3. Software that presents a unified data view while maintaining synchronization across disparate databases.

▶ Show answer

4. Which web service standard relies exclusively on XML and WSDL for strict contract enforcement?

▶ Show answer

5. Step 5 of the 8 steps, which implements Single Sign-On and unified access policies.

▶ Show answer

6. The unmanaged, continuous changes to the parameters of the original project goals.

▶ Show answer

7. The group of senior management and key users that authorizes scope modifications.

▶ Show answer

8. The ERP life cycle stage where Gap Analysis and the sandbox/prototype are done.

▶ Show answer

9. The implementation strategy that uses out-of-the-box core ERP functionality and adapts processes to the software.

▶ Show answer

10. The cutover method that runs the legacy and new systems concurrently.

▶ Show answer

11. Enumerate the 5 key planning questions.

▶ Show answer

12. Enumerate the 5 stages of the ERP life cycle in order.

▶ Show answer

B. Multiple choice

1. Which ERP generation used multi-tenant, cloud-hosted systems with microservices?
A. 1st B. 2nd C. 3rd D. None

▶ Show answer

2. Which is a characteristic of REST?
A. XML only B. Stateless, uses HTTP methods, usually JSON C. Built-in WS-Security D. Requires WSDL

▶ Show answer

3. Which interface is best suited for high-security banking and ERP transactions?
A. REST B. SFTP C. SOAP D. AMQP

▶ Show answer

4. Converting CRM status "Pending_Approval" into ERP "Code_02" is an example of:
A. Data type casting B. Value mapping C. Aggregation D. Encapsulation

▶ Show answer

5. Which step of the 8-step framework defines long-term procurement guidelines?
A. Step 4 B. Step 6 C. Step 7 D. Step 8

▶ Show answer

6. Scope creep directly affects which constraints?
A. Cost, Time, Quality B. Risk, Scope, People C. Cost, Risk, Vendor D. Time, Staff, Hardware

▶ Show answer

7. Which is NOT part of the change control white paper?
A. Issue identification B. Options and trade-offs C. Resource impact D. Vendor contract renewal

▶ Show answer

8. The primary goal of ERPLC compared to SDLC is to:
A. Build software from scratch B. Align business processes with a prepackaged platform C. Maximize code customization D. Avoid training

▶ Show answer

9. Which strategy is the most expensive, lengthy and high-risk?
A. Vanilla B. Middle-of-the-Road C. Comprehensive D. Pilot

▶ Show answer

10. Which cutover has the highest upfront cost but the lowest operational risk?
A. Direct B. Parallel C. Phased D. Pilot

▶ Show answer

C. True or false (analysis)

1. Physical integration alone guarantees enterprise success.

▶ Show answer

2. A custom OMS endpoint that takes account and order data in one request reduces transaction risk but increases upfront development time.

▶ Show answer

3. Strict standardization can restrict localized customization within departments.

▶ Show answer

4. External consultants should keep their knowledge to themselves to stay valuable to the project.

▶ Show answer

5. ROI from enterprise integration is usually immediate and fully tangible.

▶ Show answer

6. Heavy customization of core ERP code is preferred over BPR.

▶ Show answer

7. Without structured decision governance, stakeholders may not properly evaluate trade-offs, leading to a lack of user buy-in.

▶ Show answer

8. Executive sponsorship is needed only at go-live.

▶ Show answer